

Incremental Novel Class Discovery Experiments

Metrics & Terminology

1. Known-Class Accuracy (KCA):
$$KCA = (\text{Correct predictions on known classes}) / (\text{Total samples of known classes})$$
2. Forgetting = (Best past accuracy – Current accuracy), averaged over all learned classes
Example:
If dog accuracy was 0.76 earlier and becomes 0.21 later:
→ Forgetting = $0.76 - 0.21 = 0.55$
3. Average Forgetting = mean forgetting over all learned classes

Experiment 1: Single Novel Class (Dog)

Setup

Task 0: Supervised training on airplane and automobile (10,000 images).

Tasks 1–5: Stream of unseen class (dog), 1000 images per task, no labels.

Model performs novelty detection, buffering, clustering (HDBSCAN), and promotion.

Task-wise Metrics

Task	KCA	Avg. Forgetting	Event
0	0.988	0.000	Initialisation
1	0.988	0.000	Buffering
2	0.988	0.000	Buffering
3	0.866	0.093	Dog promoted
4	0.866	0.062	Stable
5	0.866	0.062	Stable

Class-wise Accuracy Trend

Task	airplane	automobile	dog
0	0.989	0.987	-
1	0.989	0.987	-
2	0.989	0.987	-
3	0.885	0.905	0.808
4	0.885	0.905	0.808
5	0.885	0.905	0.808

Purity Table

Task	ClusterID	Dominant Class	Purity	Cluster Size	Status
3	0	dog	1.00	192	Promoted

Experiment 1 (Single Novel Class)

The model handles one new class quite well because the data is simple and comes in gradually. Since there is no confusion with other new classes, clustering works cleanly and the old classes are mostly preserved. The drop is simply because of realignment of the boundaries.

Experiment 2: Sequential Classes

Setup

Task 0: Supervised initialization.

Tasks 1–40: Sequential classes (5 tasks each, 1000 images/task, no labels).

Order: bird → cat → deer → dog → frog → horse → ship → truck.

Buffer cleared after each class phase.

Key Metrics

Task	KCA	Avg. Forgetting	Event
0-17	0.988	0.000	No promotion
18	0.863	0.076	Dog promotion
21	0.714	0.256	Frog promotion
26	0.725	0.202	Horse promotion
31	0.739	0.166	Ship promotion
37	0.729	0.128	Truck promotion

Class-wise Accuracy Trend

Task	airplane	automobile	dog	frog	horse	ship	truck
0	0.989	0.987	-	-	-	-	-
18	0.907	0.916	0.766	-	-	-	-
21	0.860	0.898	0.215	0.883	-	-	-
26	0.799	0.904	0.249	0.865	0.809	-	-
31	0.733	0.882	0.375	0.838	0.774	0.831	-
37	0.682	0.862	0.497	0.859	0.751	0.845	0.604

Purity Tables

Promoted Clusters

Class	Task	ClusterID	Purity	Size
dog	18	0	1.00	192
frog	21	0	1.00	250
horse	26	2	1.00	224
ship	31	1	1.00	319

truck	37	0	1.00	535
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Experiment 2 (Sequential Classes)

Even though each new class is learned correctly, the model slowly forgets older classes over time. This happens because every new class slightly changes the internal representation, which affects what was learned earlier.

Experiment 3: Mixed Open-World Setting

Setup

Task 0: Supervised initialization.

Tasks 1–40: All 10 classes mixed (~1250 samples/task), no labels.

Key Metrics

Task	KCA	Avg. Forgetting	Event
0	0.988	0.000	Initialisation
2	0.896	0.082	Frog promotion
3	0.713	0.249	Deer promotion
4	0.530	0.402	Bird and Truck both promoted

Class-wise Accuracy Trend

Task	airplane	automobile	frog	deer	bird	truck
0	0.989	0.987	-	-	-	-
2	0.894	0.918	0.875	-	-	-
3	0.735	0.908	0.462	0.747	-	-
4	0.710	0.749	0.230	0.300	0.493	0.701

Purity Table

Class	Task	ClusterID	Purity	Size
frog	2	1	0.17	166
deer	3	1	0.17	345
bird	4	0	0.15	321
Truck	4	1	0.32	152

Experiment 3 (Mixed Setting)

Here everything breaks because all classes are mixed together from the start. The model is not able to form clean clusters, so it ends up learning incorrect groupings and overall performance drops a lot.

Experiment 3 (Modified setting – open world with introduction of 2-3 classes at a time)

Stream setup 1:

In this method, new classes are completely exhausted (all 5,000 images shown) within their specific phase.

After their phase ends, they become "reminders" (100 reused images per task) alongside Task 0 classes.

Task 0 (Initialization)

Class 0 (Airplane): 5,000 images

Class 1 (Automobile): 5,000 images

Total Batch Size: 10,000

Phase 1 (Tasks 1 to 13) - Exhausting Class 2 & Class 3

Class 0, Class 1 (Reminders): 100 images each (Reused)

Class 2, Class 3 (New Wave): ~384 images each (New) (5,000 imgs / 13 tasks)

Total Batch Size per Task: ~968 images

Phase 2 (Tasks 14 to 26) - Exhausting Class 4, Class 5, Class 6

Class 0, Class 1 (Reminders): 100 images each (Reused)

Class 2, Class 3 (Reminders): 100 images each (Reused from Phase 1)

Class 4, Class 5, Class 6 (New Wave): ~384 images each (New) (5,000 imgs / 13 tasks)

Total Batch Size per Task: ~1,552 images

Phase 3 (Tasks 27 to 39) - Exhausting Class 7, Class 8, Class 9

Class 0 to Class 6 (Reminders): 100 images each (Reused) -> 700 images total

Class 7, Class 8, Class 9 (New Wave): ~384 images each (New) (5,000 imgs / 13 tasks)

Total Batch Size per Task: ~1,852 images

Results:

Key Metrics

Task	KCA	Avg. Forgetting	Event
0-4	0.992	0.000	Initialisation
5	0.917	0.078	Cat promotion
6	0.706	0.222	Bird promotion
15	0.509	0.345	Deer & Frog promotion
27	0.489	0.278	Truck & Ship promotion
39	0.489	0.208	End of stream

Class-wise Accuracy Trend

Task	airplane	automobile	cat	bird	deer	frog	truck	ship
0	0.994	0.989	-	-	-	-	-	-
5	0.863	0.964	0.923	-	-	-	-	-
6	0.843	0.945	0.452	0.583	-	-	-	-
15	0.596	0.951	0.102	0.462	0.241	0.705	-	-
27	0.539	0.894	0.288	0.440	0.098	0.510	0.554	0.588
39	0.539	0.894	0.288	0.440	0.098	0.510	0.554	0.588

Purity Table

Class	Task	ClusterID	Purity	Size
cat	5	0	0.60	1832
bird	6	2	0.54	292
deer	15	0	0.41	263
frog	15	2	0.51	207

truck	27	0	0.60	205
ship	27	1	0.37	284

Stream setup 2:

In this method, we do not exhaust the new classes immediately. Instead, we take their 5,000 images and spread them evenly across all remaining tasks until Task 39.

Task 0 (Initialization)

Class 0 (Airplane): 5,000 images

Class 1 (Automobile): 5,000 images

Total Batch Size: 10,000

Phase 1 (Tasks 1 to 13) - Starting Class 2 & Class 3

Class 0, Class 1 (Reminders): 100 images each (Reused)

Class 2, Class 3 (New Stream): ~128 images each (New) (5,000 imgs spread over 39 tasks)

Total Batch Size per Task: ~456 images

Phase 2 (Tasks 14 to 26) - Starting Class 4, Class 5, Class 6

Class 0, Class 1 (Reminders): 100 images each (Reused)

Class 2, Class 3 (Continuing Stream): ~128 images each (New)

Class 4, Class 5, Class 6 (New Stream): ~192 images each (New) (5,000 imgs spread over 26 tasks)

Total Batch Size per Task: ~1,032 images

Phase 3 (Tasks 27 to 39) - Starting Class 7, Class 8, Class 9

Class 0, Class 1 (Reminders): 100 images each (Reused)

Class 2, Class 3 (Continuing Stream): ~128 images each (New)

Class 4, Class 5, Class 6 (Continuing Stream): ~192 images each (New)

Class 7, Class 8, Class 9 (New Stream): ~384 images each (New) (5,000 imgs spread over 13 tasks)

Total Batch Size per Task: ~2,184 images

Results

Key Metrics

Task	KCA	Avg. Forgetting	Event
0-1	0.992	0.000	Initialisation
2	0.902	0.078	Cat promotion
4	0.681	0.194	Bird promotion
18	0.603	0.206	Frog promotion
27	0.573	0.197	Truck promotion
33	0.518	0.218	Ship promotion
39	0.518	0.187	End of stream

Class-wise Accuracy Trend

Task	airplane	automobile	cat	bird	frog	truck	ship
0	0.994	0.989	-	-	-	-	-
2	0.883	0.945	0.877	-	-	-	-
4	0.792	0.949	0.536	0.449	-	-	-
18	0.842	0.897	0.448	0.297	0.532	-	-
27	0.792	0.831	0.464	0.374	0.394	0.585	-
33	0.788	0.813	0.277	0.402	0.308	0.529	0.510
39	0.788	0.813	0.277	0.402	0.308	0.529	0.510

Purity Table

Class	Task	ClusterID	Purity	Size
cat	2	1	0.53	318
bird	4	1	0.39	212
frog	18	0	0.35	241
truck	27	1	0.57	229
ship	33	0	0.27	278